MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 03 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. Three unbiased coins are tossed together. The probability of getting exactly two tails is: **[1]**

Options: (a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{5}{8}$

2. Two dice are thrown simultaneously. The probability of getting a doublet is: **[1]**

Options: (a) $\frac{1}{36}$ (b) $\frac{1}{6}$ (c) $\frac{5}{6}$ (d) $\frac{1}{12}$

3. A bag contains 5 red balls and n green balls. If the probability of drawing a green ball is three times that of drawing a red ball, then the value of n is: **[1]**

Options: (a) 10 (b) 15 (c) 20 (d) 18

4. The probability that a leap year chosen at random has 53 Sundays is: **[1]**

Options: (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$

5. **Assertion (A):** In a single throw of a die, the probability of getting an even prime number is $\frac{1}{6}$.
Reason (R): The only even prime number on a standard die is 2, which is 1 favourable outcome out of 6. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. One card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability of getting: (i) an ace (ii) a black king. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. Two dice are thrown together. Find the probability that: (i) the product of the numbers is 12 (ii) the product of the numbers is odd (iii) the product of the numbers is even. **[3]**

8. A bag contains 15 balls of which x are blue and the remaining are red. (i) If a ball is drawn at random, write the expression for the probability that it is a blue ball. (ii) If 5 more blue balls are added to the bag, the probability of drawing a blue ball becomes double that in part (i). Find the value of x. **[3]**

9. Cards numbered from 1 to 49 are placed in a box and mixed thoroughly. A card is drawn at random from the box. Find the probability that the number on the card is: (i) an odd number (ii) a perfect square (iii) a multiple of 7. **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

In a national archery championship, an Olympic target is designed with five concentric scoring circles marked from the center outwards as Gold, Red, Blue, Black, and White. The diameter of the central Gold region is 21 cm and each of the four surrounding concentric bands has a uniform width of 10.5 cm. An archer releases an arrow that hits the target at random.

Based on the above scenario, answer the following:

(i) Find the radius of the central Gold region and the total radius of the entire circular board. **[1]**

(ii) What is the probability that the arrow hits the Gold region? **[1]**

(iii) Find the probability that the arrow hits the Red ring. **[1]**

(iv) What is the probability that the arrow does NOT hit the Gold region? **[1]**

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